Poynting vector and Problems on Uniform plane waves

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POYNTING'S THEOREM AND WAVE POWER

- Power flow associated with an electromagnetic wave
- Consider Maxwell's curl equation - assume that the medium may be conductive

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

- Take the scalar product of both sides with **E** $\mathbf{E} \cdot \nabla \times \mathbf{H} = \mathbf{E} \cdot \mathbf{J} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$ **---(1)**

- Introduce the following vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{E} \cdot \nabla \times \mathbf{H} + \mathbf{H} \cdot \nabla \times \mathbf{E}$

- Rewriting eq. (1) → $\mathbf{H} \cdot \nabla \times \mathbf{E} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{J} \cdot \mathbf{E} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$

- Consider $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

- Therefore $-\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{J} \cdot \mathbf{E} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}$

Rewriting…

$$-\mathbf{H}\cdot\frac{\partial \mathbf{B}}{\partial t}-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\mathbf{E}\cdot\frac{\partial \mathbf{D}}{\partial t}$$

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\epsilon\mathbf{E}\cdot\frac{\partial \mathbf{E}}{\partial t}+\mu\mathbf{H}\cdot\frac{\partial \mathbf{H}}{\partial t}$$

- But $\epsilon\mathbf{E}\cdot\frac{\partial \mathbf{E}}{\partial t}=\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)$ (Rearranging the derivatives)

$$\mu\mathbf{H}\cdot\frac{\partial \mathbf{H}}{\partial t}=\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)$$

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H})=\mathbf{J}\cdot\mathbf{E}+\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)+\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)$$

$$-\int_{\text{vol}}\nabla\cdot(\mathbf{E}\times\mathbf{H})\,dv=\int_{\text{vol}}\mathbf{J}\cdot\mathbf{E}\,dv+\int_{\text{vol}}\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\right)dv+\int_{\text{vol}}\frac{\partial}{\partial t}\left(\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\right)dv$$

$$-\oint_{\text{area}}(\mathbf{E}\times\mathbf{H})\cdot d\mathbf{S}=\int_{\text{vol}}\mathbf{J}\cdot\mathbf{E}\,dv+\frac{d}{dt}\int_{\text{vol}}\frac{1}{2}\mathbf{D}\cdot\mathbf{E}\,dv+\frac{d}{dt}\int_{\text{vol}}\frac{1}{2}\mathbf{B}\cdot\mathbf{H}\,dv$$

Poynting's theorem

$$-\oint_{\text{area}} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = \int_{\text{vol}} \mathbf{J} \cdot \mathbf{E}\, dv + \frac{d}{dt} \int_{\text{vol}} \frac{1}{2} \mathbf{D} \cdot \mathbf{E}\, dv + \frac{d}{dt} \int_{\text{vol}} \frac{1}{2} \mathbf{B} \cdot \mathbf{H}\, dv$$

Total power flowing *out* of the volume

Ohmic power dissipated within the volume

Total energy stored in the electric field

Total energy stored in the magnetic field

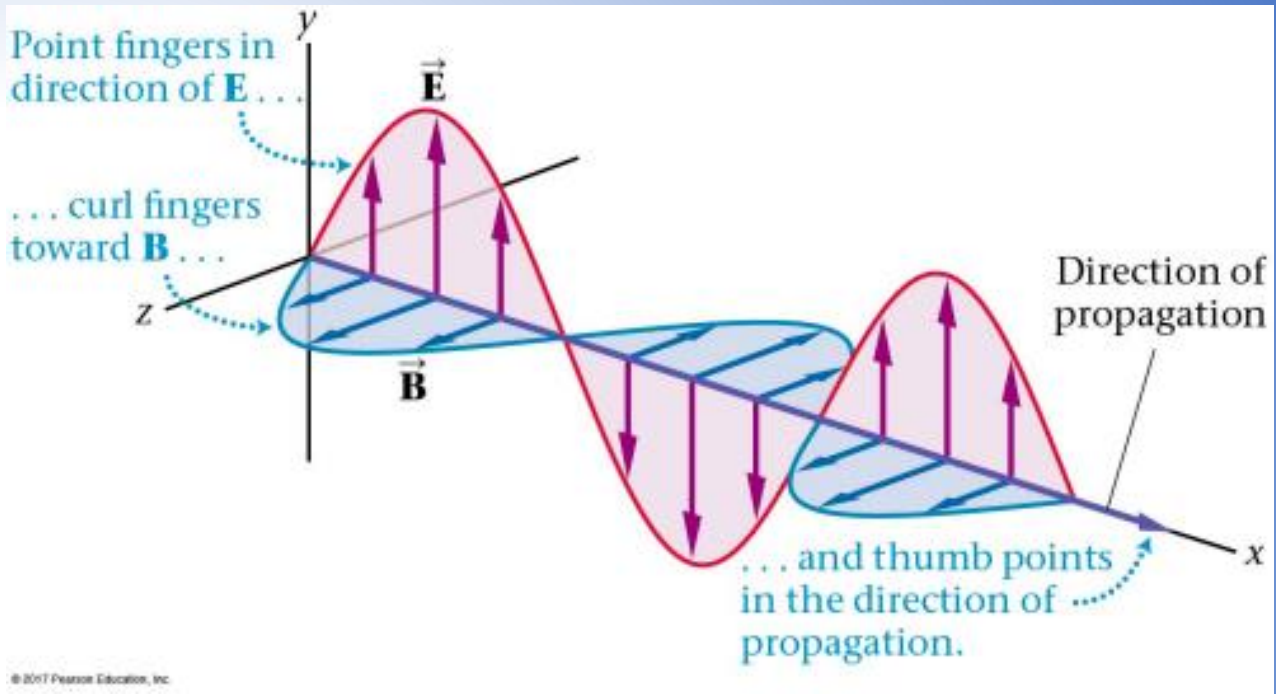
$$\oint_{\text{area}} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} \quad \text{W}$$

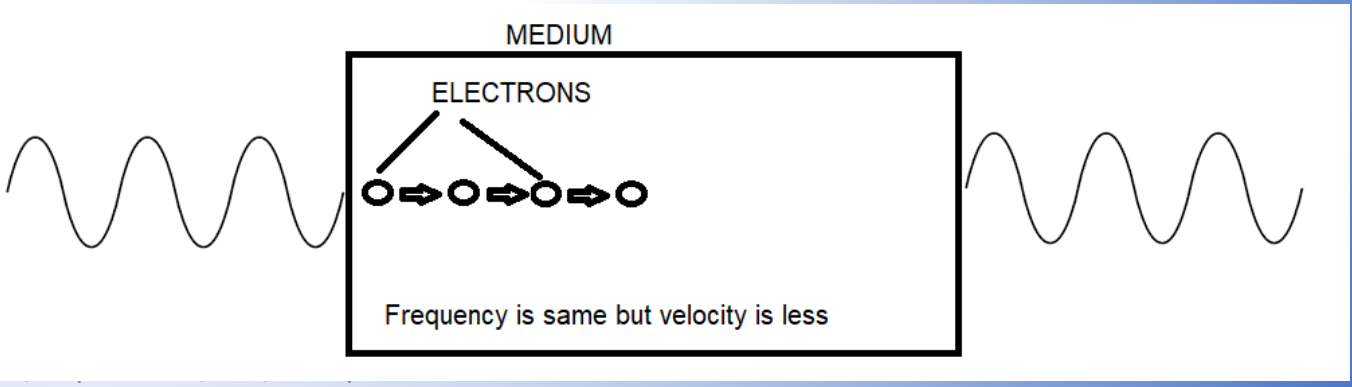
Poynting vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad \text{W/m}^2$$

instantaneous power density

- EM waves “propagate” – Source vibrates at a frequency
- This is carried by EM wave – source not required – self propagation
- EM wave don’t require medium for propagation
- If it hits a media – media carries the vibration
- Freq. don’t change, amplitude may change
- Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad \mathrm{W/m^2}$
- Gives instantaneous power density





Revision

Uniform plane wave propagation in free space:

$$\mathbf{E}(z, t) = E_o \cos(\omega t - \beta z)\mathbf{a}_x$$

$$\lambda = \frac{2\pi}{\beta} \qquad \beta = \omega\sqrt{\mu_0\varepsilon_0} \qquad c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \qquad \eta = \sqrt{\frac{\mu_0}{\varepsilon_0}} \qquad H_o = \frac{E_o}{\eta}$$

Uniform plane wave propagation in partially conducting medium (Lossy dielectric)

$$\mathbf{E}(z, t) = E_o e^{-\alpha z}\cos(\omega t - \beta z)\mathbf{a}_x$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$\mathbf{k} = \alpha + j\beta$ $\mathbf{k}$, α, and β are propagation constant, attenuation constant and phase constant respectively

$$\alpha = \mathrm{Re}\{jk\} = \omega\sqrt{\frac{\mu\epsilon'}{2}}\left(\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} - 1\right)^{1/2}$$

$$\beta = \mathrm{Im}\{jk\} = \omega\sqrt{\frac{\mu\epsilon'}{2}}\left(\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} + 1\right)^{1/2}$$

$\varepsilon^{|} = \varepsilon$ *and* $\varepsilon^{||} = \sigma/\omega$

Uniform plane wave propagation in perfect dielectric (Lossless dielectric) medium:

$$\beta = \omega\sqrt{\mu\,\varepsilon} \qquad \eta = \sqrt{\frac{\mu}{\varepsilon}} \qquad v = \frac{1}{\sqrt{\mu\,\varepsilon}}$$

Uniform plane wave propagation in a good conductor:

Skin depth $$\delta = \frac{1}{\sqrt{\pi f\,\mu\sigma}} = \frac{1}{\alpha} = \frac{1}{\beta}$$

and $$\eta = \sqrt{\frac{\omega\mu}{\sigma}}\,.\,e^{\pi/4}$$

An **H** field in free space is given as $\mathbf{H}(x,t) = 10\cos(10^8 t - \beta x)\mathbf{a}_y$ A/m.
Find: (a) β; (b) λ; (c) $\mathbf{E}(x,t)$ at $P(0.1, 0.2, 0.3)$ at $t = 1$ ns.

$\beta = \omega/c$, $\beta = 10^8/(3 \times 10^8) = \underline{0.33 \text{ rad/m}}$.

b) λ: We know $\lambda = 2\pi/\beta = \underline{18.9 \text{ m}}$.

$$E(x,t) = -\eta_0 H(x,t) = -3.77 \times 10^3 \cos(10^8 t - \beta x)$$

The vector direction of **E** will be $-\mathbf{a}_z$,

$-a_z \times a_y = a_x$

Direction of propagation is E X H

At $x = 0.1$, $t = 10^{-9}$ sec,

$$\mathbf{E}(x,t) = -3.77 \times 10^3 \cos[(10^8)(10^{-9}) - (0.33)(0.1)]\mathbf{a}_z$$

$$= -3.77 \times 10^3 \cos(6.7 \times 10^{-2})\mathbf{a}_z$$

$$= \underline{-3.76 \times 10^3 \mathbf{a}_z \text{ V/m}}$$

2. A wave propagating in lossless dielectric has the components E = 500 cos (10^7 t-βz) a_x V/m and H = 1.1 cos (10^7 t-βz) a_y. If the wave is travelling at v = 0.5c, find
a) μ_r b) ε_r c) β d) λ e) η

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

$$V = \frac{1}{\sqrt{\mu\varepsilon}}$$

Given $\omega = 10^7$, $E_0 = 500$ V/m, $H_0 = 1.1$ A/m, $v = 0.5c = 1.5 \times 10^8$ m/s

a) $\eta = \frac{E_0}{H_0} = \sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}} = \frac{500}{1.1} \Rightarrow \frac{\mu_r}{\epsilon_r} = 1.455$

$v = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}} = 1.5 \times 10^8 \Rightarrow \mu_r\epsilon_r = 4$

Solving we get $\mu_r = \sqrt{5.823}$, $\epsilon_r = \sqrt{2.413}$

$$\beta = \omega\sqrt{\mu\varepsilon}$$

c) $\beta = \omega\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}$

=0.0645 rad/m

d) $\lambda = \frac{2\pi}{\beta} = 94.2$ m

e) $\eta = \frac{500}{1.1} = 454.545\ \Omega$

Pb. Write the time domain expression for the electric field of a uniform plane wave in a lossy medium ($\mu_r = 1$) if

$$\vec{H} = 0.1\, e^{-200y} \cos(2\pi \times 10^{10} t - 300y)\, \hat{a}_x \ (A/m)$$

Solution: Given $\omega = 2\pi \times 10^{10}$ rad/sec

$\beta = 300$

$H_0 = 0.1$

$\alpha = 200$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

$$k = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$200 + j300 = \sqrt{j2\pi \times 10^{10} \mu_0 (\sigma + j2\pi \times 10^{10} \epsilon_0 \epsilon_r)}$$

$$(200 + j300)^2 = -\frac{16}{36}\pi^2 \times 10^4 \epsilon_r + j2\pi \times 10^{10} \times 4\pi \times 10^{-7} \sigma$$

$$(200 + j300)^2 = -\frac{16}{36}\pi^2 \times 10^4 \epsilon_r + j2\pi \times 10^{10} \times 4\pi \times 10^{-7}\sigma$$

Equating the real and imaginary parts we get

$$\epsilon_r = 1.14 \quad \text{and} \quad \sigma = 1.52 \ (S/m)$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j\omega\mu_0\mu_r}{\sigma + j\omega\epsilon_0\epsilon_r}} = 219\angle 33.7^\circ \ (\Omega) = \frac{E_0}{H_0}$$

$$E = 0.1 \times 219 \times e^{-200y} \cdot \cos(2\pi \times 10^{10} t - 300y + 33.7^\circ)\ \hat{a}_2$$

$a_z \times a_x = a_y$

A good conductor is planar in form and carries a uniform plane wave that has a wavelength of 0.3 mm and a velocity of 3×10^5 m/s. Assuming the conductor is non-magnetic, determine the frequency and the conductivity:

$$f = \frac{v}{\lambda} = \frac{3 \times 10^5}{3 \times 10^{-4}} = 10^9 \text{ Hz} = \underline{1 \text{ GHz}}$$

$\delta = \frac{1}{\alpha} = \frac{1}{\beta}$ and $\lambda = \frac{2\pi}{\beta} = 2\pi\delta$

$$\delta = \frac{\lambda}{2\pi} = \frac{1}{\sqrt{\pi f \mu \sigma}} \quad \Rightarrow \quad \sigma = \frac{4\pi}{\lambda^2 f \mu} = \frac{4\pi}{(9 \times 10^{-8})(10^9)(4\pi \times 10^{-7})} = \underline{1.1 \times 10^5 \text{ S/m}}$$


Thank
you